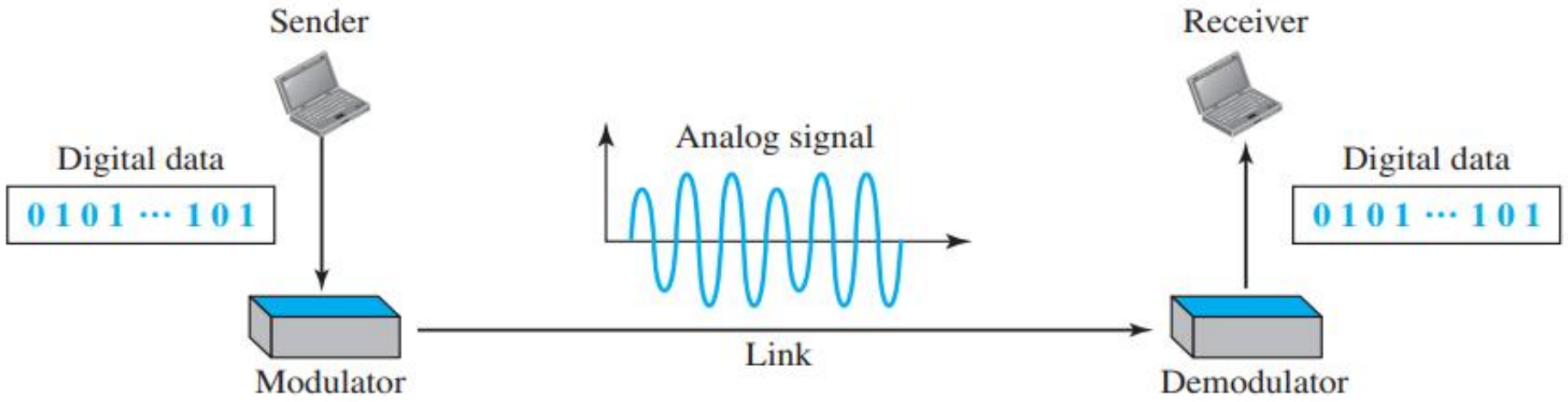
Digital to Analog Convertor

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- Phase Shift Keying (PSK)
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Introduction

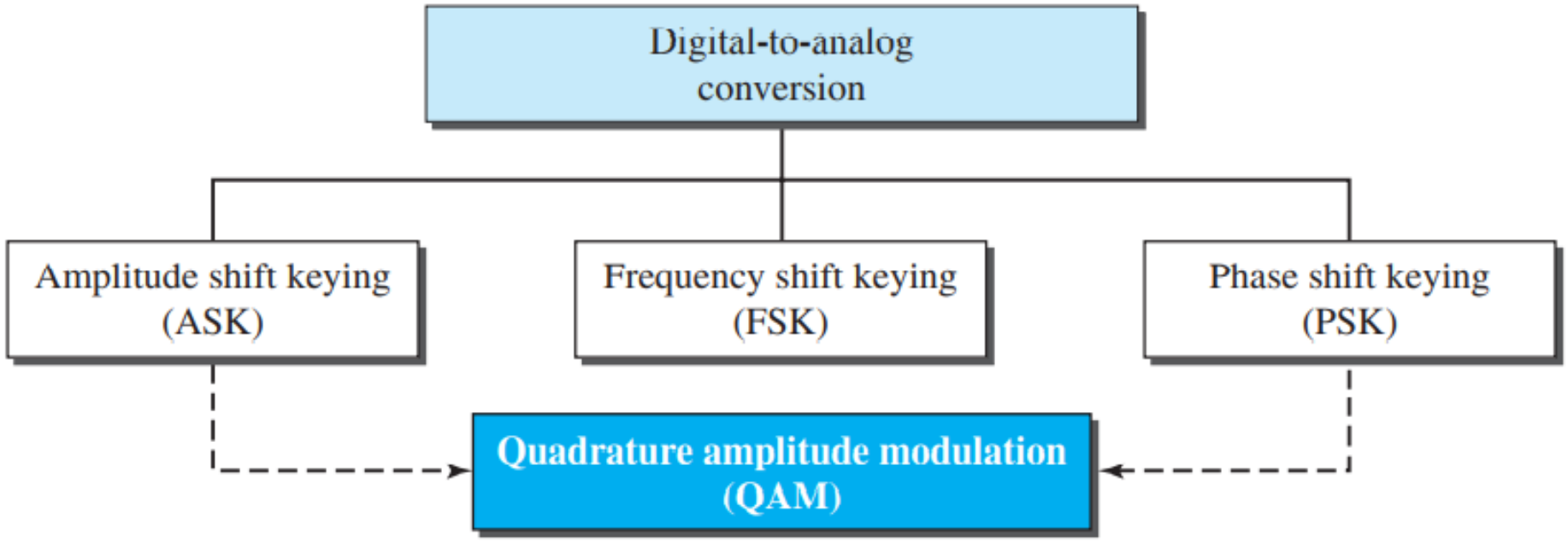
- Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data.



Contd…

- Three mechanisms for modulating digital data into an analog signal:
  - Amplitude shift keying (ASK)
  - Frequency shift keying (FSK)
  - Phase shift keying (PSK)
- Mechanism that combines changing both the amplitude and phase, called quadrature amplitude modulation (QAM).
- QAM is the most efficient and the mechanism commonly used today.

Types of Digital-to-Analog conversion



Contd…

- Bit rate is the number of bits per second.
- Baud rate is the number of signal elements per second.
- In the analog transmission of digital data, the baud rate is less than or equal to the bit rate

Activity - Concept Check

- An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate.

$S = N \times (1/r)$ $\Rightarrow$ $N = S \times r$

$= 1000 \times 4$

$N = 4000$ bps

Activity - Concept Check

- An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

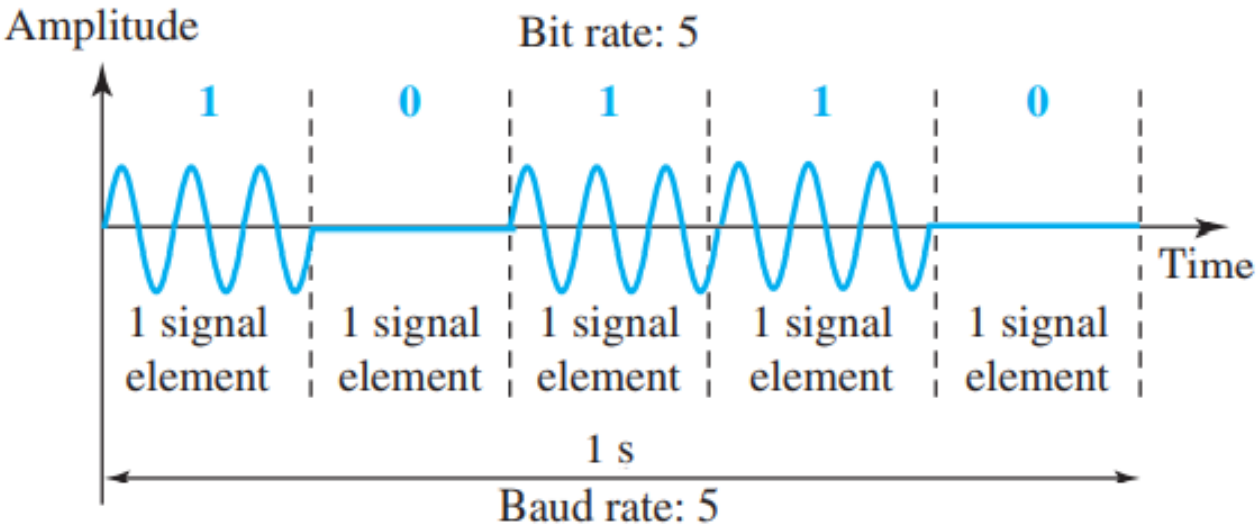
$S = N \times (1/r)$ $\quad\quad$ $r = N / S$

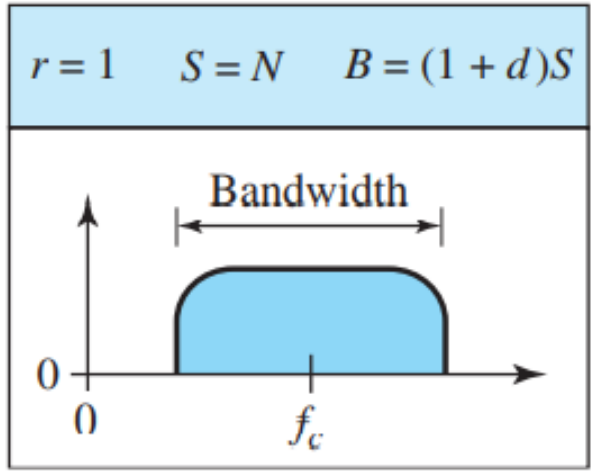
$r = \log_2 L$

Amplitude Shift Keying

- In amplitude shift keying, the amplitude of the carrier signal is varied to create signal elements.
- Both frequency and phase remain constant while the amplitude changes.
- ASK is normally implemented using only two levels.
- This is referred to as binary amplitude shift keying or on-off keying (OOK).
- The peak amplitude of one signal level is 0; the other is the same as the amplitude of the carrier frequency

Binary amplitude shift keying

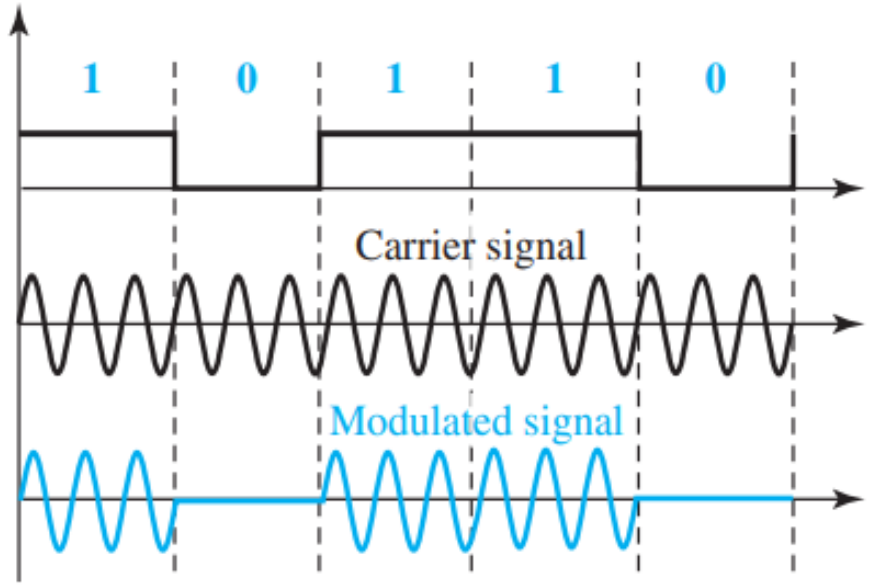


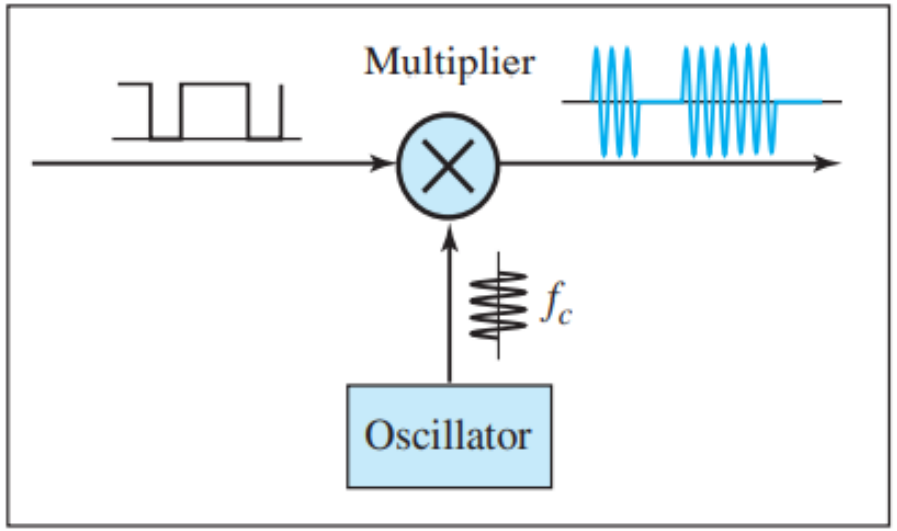


Bandwidth can be expressed as S is the signal rate and the B is the bandwidth.

$$B = (1 + d) \times S$$

Implementation





Contd…

- If digital data are presented as a unipolar NRZ digital signal with a high voltage of 1 V and a low voltage of 0 V,
- The implementation can achieved by multiplying the NRZ digital signal by the carrier signal coming from an oscillator.
- When the amplitude of the NRZ signal is 1, the amplitude of the carrier frequency is held;
- when the amplitude of the NRZ signal is 0, the amplitude of the carrier frequency is zero.

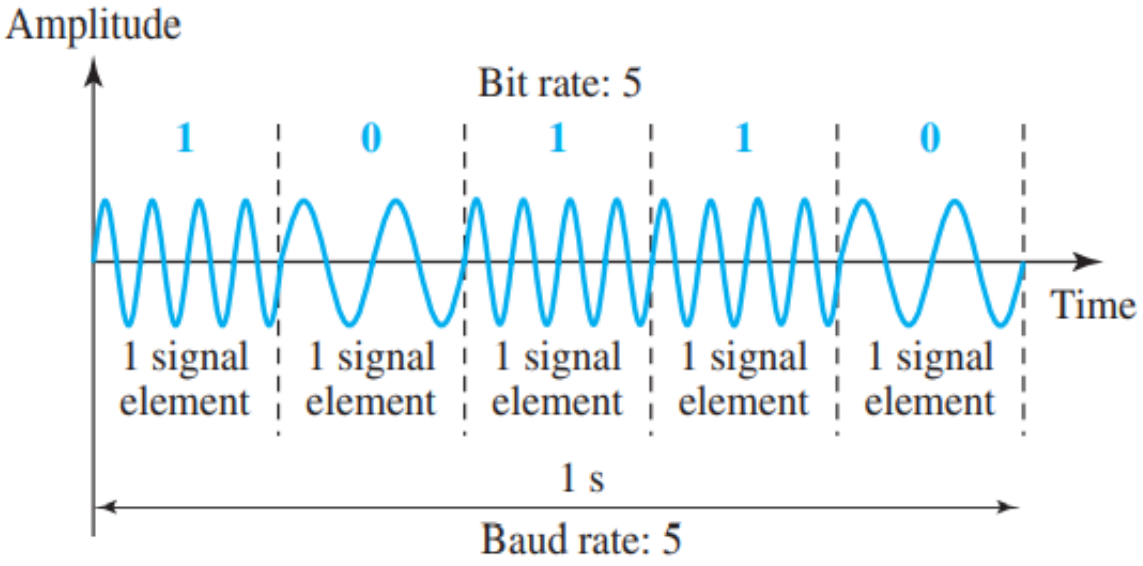
Frequency Shift Keying

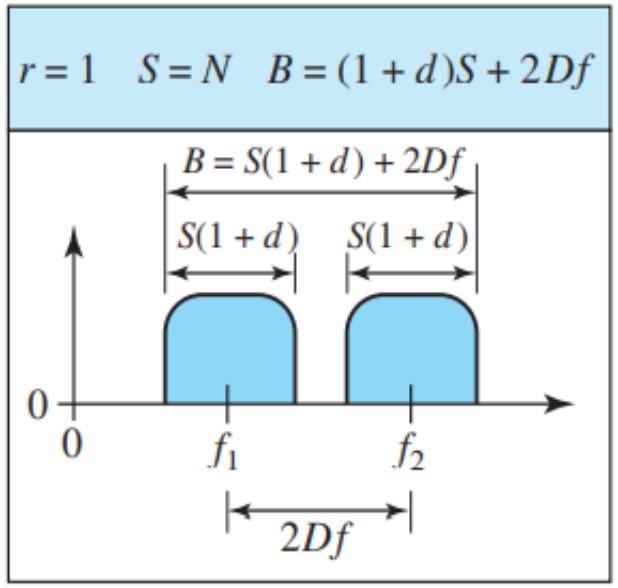
- In frequency shift keying, the frequency of the carrier signal is varied to represent data.
- The frequency of the modulated signal is constant for the duration of one signal element, but changes for the next signal element if the data element changes.
- Both peak amplitude and phase remain constant for all signal elements.

Binary FSK (BFSK)

- One way to think about binary FSK (or BFSK) is to consider two carrier frequencies.
- Two carrier frequencies, f1 and f2.
- The first carrier if the data element is 0; we use the second if the data element is 1.
- The carrier frequencies are very high, and the difference between them is very small.

Contd…





- The middle of one bandwidth is f1 and the middle of the other is f2.
- Both f1 and f2 are Δf apart from the midpoint between the two bands.
- The difference between the two frequencies is 2Δf

Bandwidth for BFSK

- FSK as two ASK signals, each with its own carrier frequency (f1 or f2).
- If the difference between the two frequencies is $2\Delta f$, then the required bandwidth is

$$B = (1 + d) \times S + 2\Delta\varphi$$

Concept Check

- We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What should be the carrier frequency and the bit rate if we modulated our data by using FSK with d = 1?

$$B = (1 + d) \times S + 2\Delta_f = 100\ \text{KHZ}$$

Choose $2\Delta_f$ to be 50 kHz

$2S = 50\text{KHZ}$

$S = 25\text{KHZ}$

250 KHZ

$S = N \times \frac{1}{r \to 1}$

$N = 25\ \text{bps}$

200 to 300 KHZ

d = 1 ?

Frequency of carrier signal

Bit rate

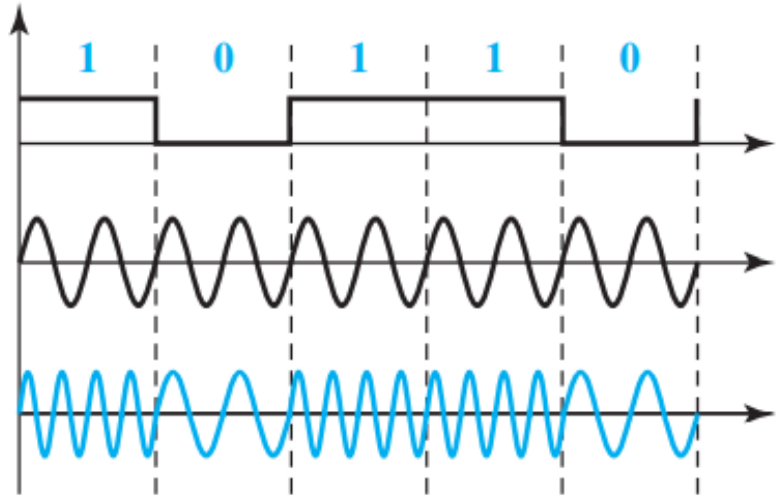
Implementation

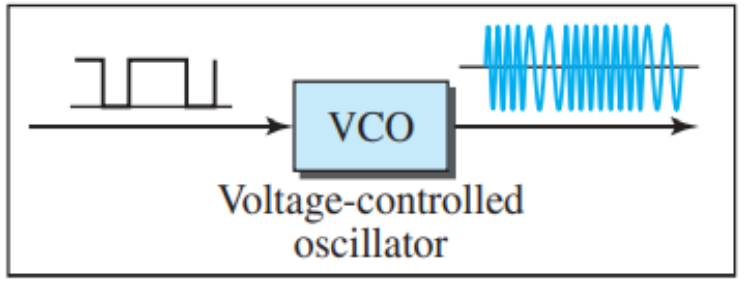
- There are two implementations of BFSK: noncoherent and coherent.
- In noncoherent BFSK, there may be discontinuity in the phase when one signal element ends and the next begins.
- In coherent BFSK, the phase continues through the boundary of two signal elements.
- Noncoherent BFSK can be implemented by treating BFSK as two ASK modulations and using two carrier frequencies.

Contd…

- Coherent BFSK can be implemented by using one voltage-controlled oscillator (VCO) that changes its frequency according to the input voltage.
- When the amplitude of NRZ is zero, the oscillator keeps its regular frequency; when the amplitude is positive, the frequency is increased.

Contd…





Multilevel FSK

- Multilevel modulation (MFSK) is not uncommon with the FSK method.
- We can use more than two frequencies.
- For example, we can use four different frequencies f_1, f_2, f_3, and f_4 to send 2 bits at a time.
- To send 3 bits at a time, eight frequencies.
- Bandwidth is

$$\mathbf{B = (1 + d) \times S + (L - 1)2D_f \rightarrow B = L \times S}$$

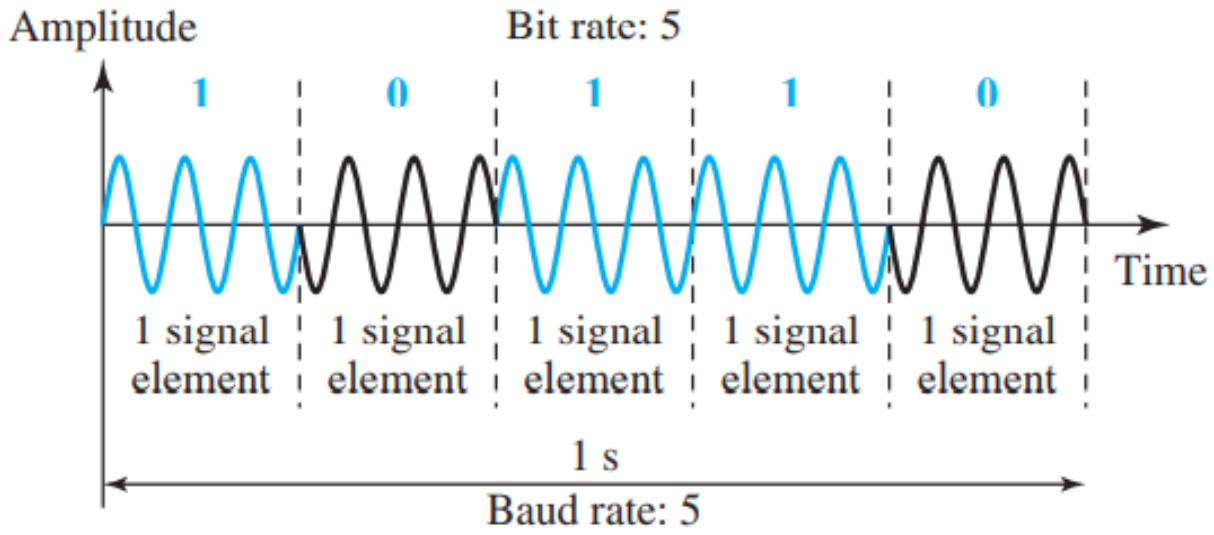
Phase Shift Keying

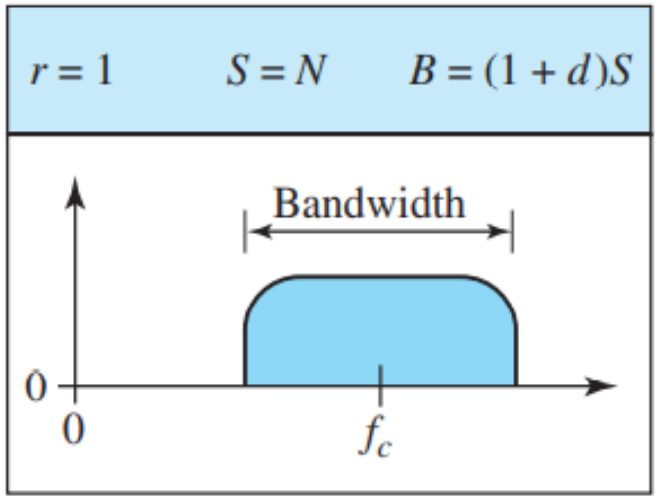
- In phase shift keying, the phase of the carrier is varied to represent two or more different signal elements.
- Both peak amplitude and frequency remain constant as the phase changes.
- Today, PSK is more common than ASK or FSK.
- QAM, which combines ASK and PSK, is the dominant method of digital-to-analog modulation.

Binary PSK (BPSK)

- The simplest PSK is binary PSK, in which we have only two signal elements, one with a phase of 0°, and the other with a phase of 180°.
- Binary PSK is as simple as binary ASK with one big advantage—it is less susceptible to noise.
- Noise can change the amplitude easier than it can change the phase. In other words, PSK is less susceptible to noise than ASK.
- PSK is superior to FSK because we do not need two carrier signals.
- PSK needs more sophisticated hardware to be able to distinguish between Phases.

Binary phase shift keying

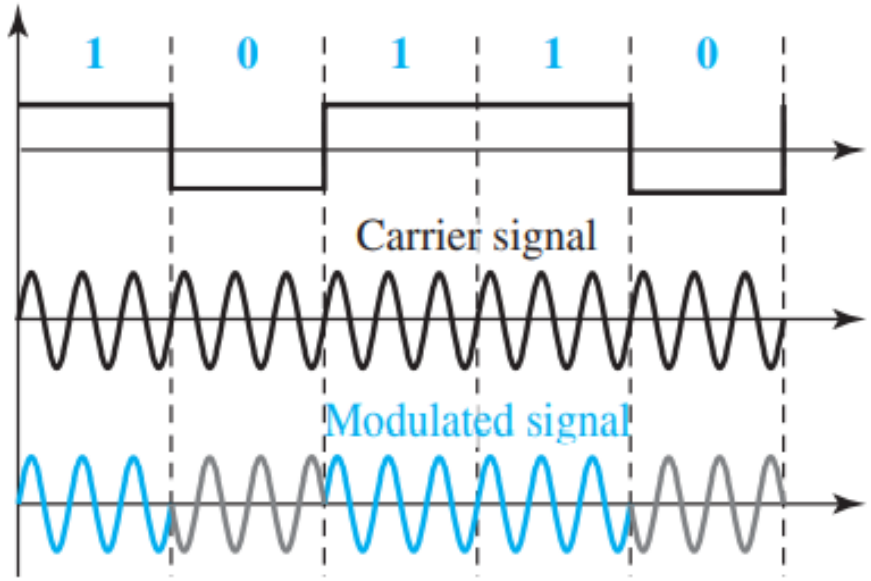


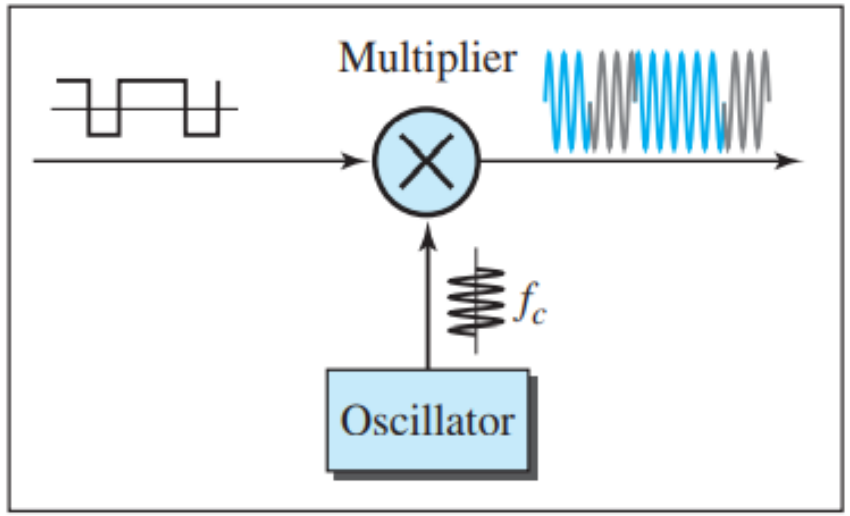


Implementation

- The reason is that the signal element with phase 180° can be seen as the complement of the signal element with phase 0°.
- The polar NRZ signal is multiplied by the carrier frequency; the 1 bit (positive voltage) is represented by a phase starting at 0°; the 0 bit (negative voltage) is represented by a phase starting at 180°.

Implementation of BASK

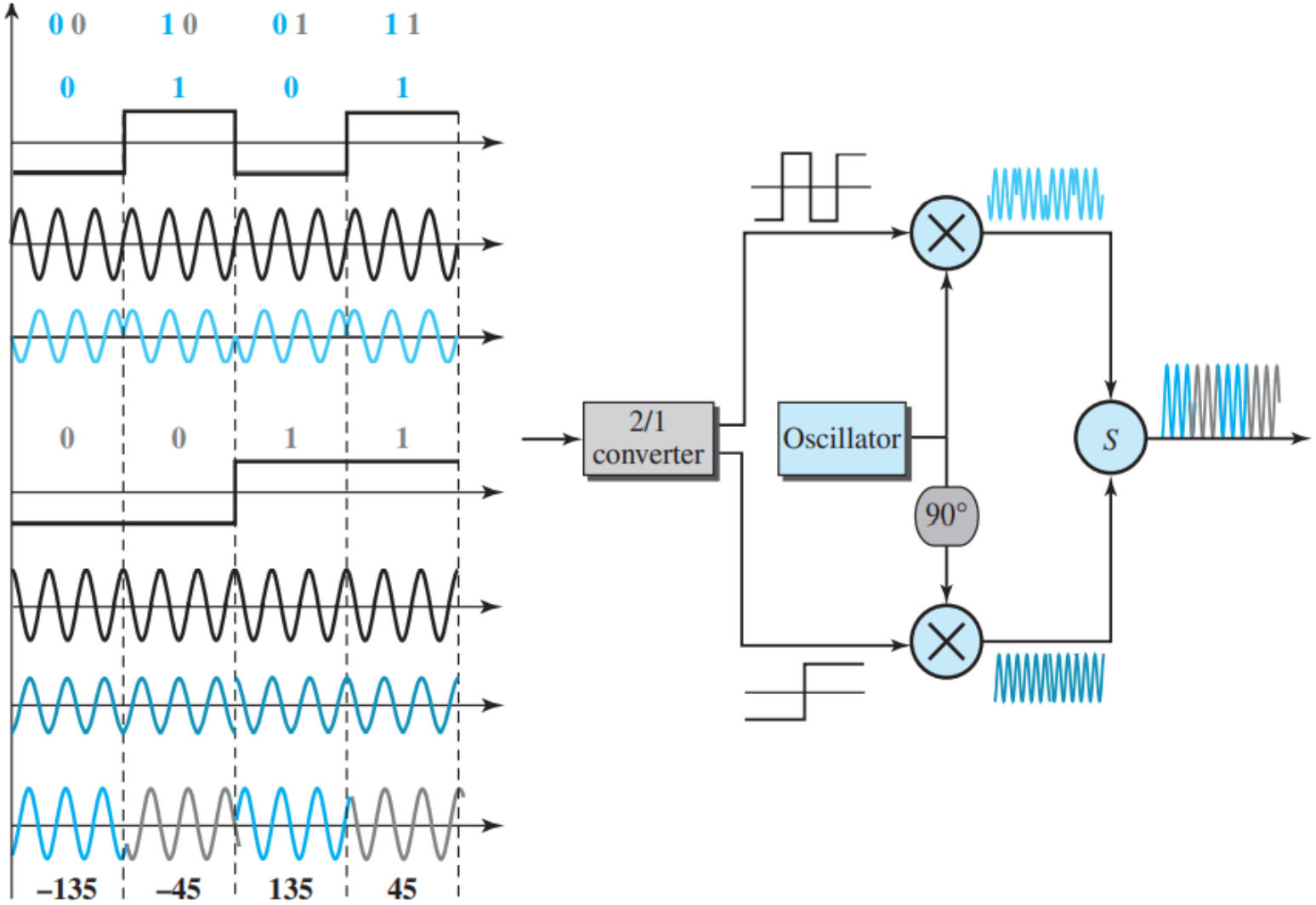




Quadrature PSK (QPSK)

- The scheme is called quadrature PSK or QPSK because it uses two separate BPSK modulations; one is in-phase, the other quadrature (out-of-phase).
- The incoming bits are first passed through a serial-to-parallel conversion that sends one bit to one modulator and the next bit to the other modulator.
- If the duration of each bit in the incoming signal is T, the duration of each bit sent to the corresponding BPSK signal is 2T.
- This means that the bit to each BPSK signal has one-half the frequency of the original signal.

QPSK and its implementation



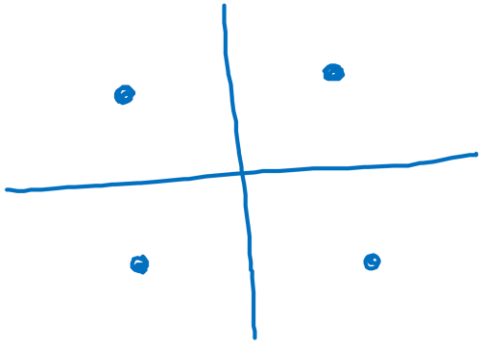
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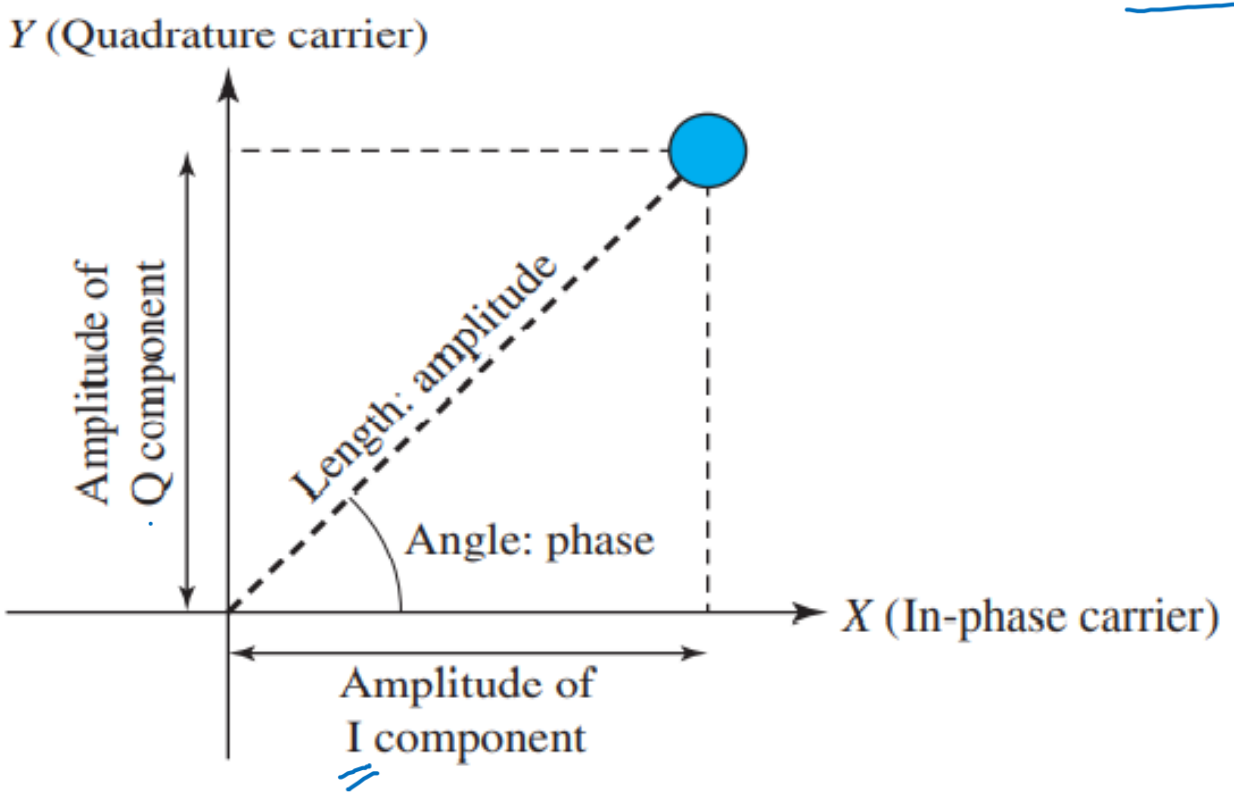
- The two composite signals created by each multiplier are sine waves with the same frequency, but different phases.
- The result is another sine wave, with one of four possible phases: 45°, −45°, 135°, and −135°.
- Four kinds of signal elements in the output signal (L = 4),
- 2 bits per signal element (r = 2) are sent.

Constellation Diagram

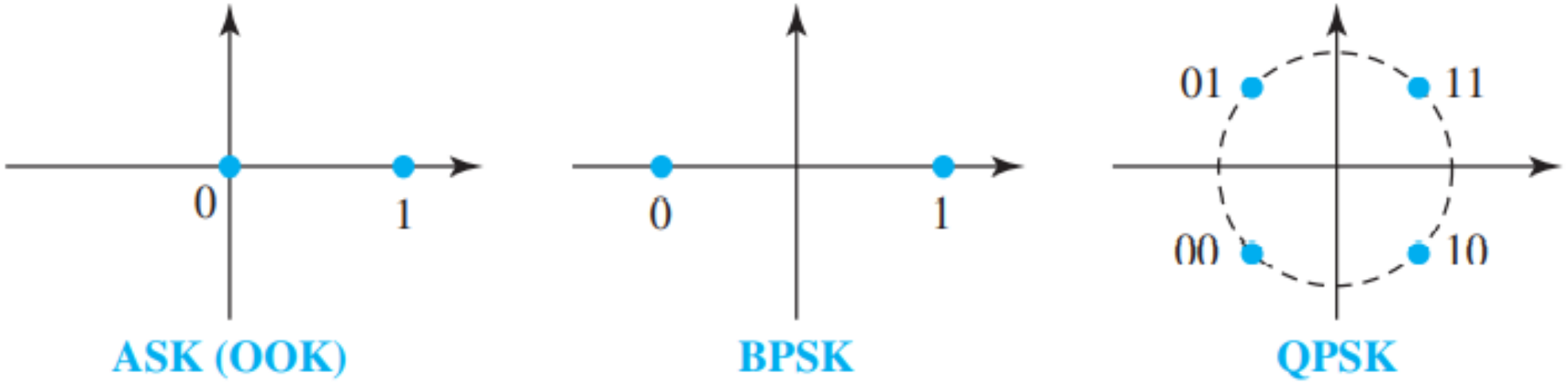
- A constellation diagram helps to define the amplitude and phase of a signal element, when two carriers is used (one in-phase and one quadrature).
- In a constellation diagram, a signal element type is represented as a dot

Concept of a constellation diagram





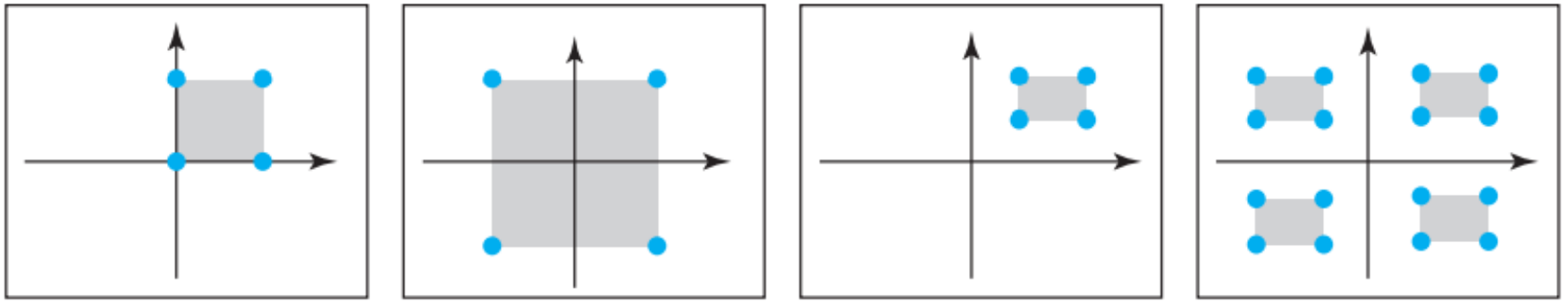
Show the constellation diagrams for ASK (OOK), BPSK, and QPSK signals.



Quadrature Amplitude Modulation

- PSK is limited by the ability of the equipment to distinguish small differences in phase. This factor limits its potential bit rate
- The idea of using two carriers, one in-phase and the other quadrature, with different amplitude levels for each carrier is the concept behind quadrature amplitude modulation (QAM).
- Quadrature amplitude modulation is a combination of ASK and PSK.

Constellation diagrams for some QAMs



a. 4-QAM b. 4-QAM c. 4-QAM d. 16-QAM

Bandwidth for QAM: The minimum bandwidth required for QAM transmission is the same as that required for ASK and PSK transmission.